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Education

2025 – present **Ph.D. Student in Cybersecurity** at Korea University.

2021 – 2025 **B.S. in Mathematics, B.E. in Smart Security** at Korea University.
- Micro Degree in Cryptographic Technology
- GPA: 3.71/4.5

Research Publications

International Journals & Conferences

1. D. Kang, H. Shin, D. Kim, S. Hong, and H. Kim, "SDDT: an operation skip attack framework for bitslice ciphers - validated on PIPO," *Selected Areas in Cryptography*, 2026.
2. H. Shin, D. Kang, D. Kwon, S. Hong, and H. Kim, "IISFA: ineffective instruction-skip fault attacks on bitslice implementations," *IACR Cryptol. ePrint Arch.*, vol. 2026, p. 2536, 2026. URL: https://eprint.iacr.org/2026/2536
3. H. Shin, D. Kang, D. Kwon, S. Hong, and H. Kim, "Sat and neural network filters for misinjections in fault attacks," *IACR Cryptol. ePrint Arch.*, vol. 2026, p. 2537, 2026. URL: https://eprint.iacr.org/2026/2537
4. D. Kang, J. Kang, and S. Lee, *Defense mechanism using purifier and detector against adversarial patch attacks for unmanned combat systems*, Poster presented at the World Conference on Information Security Applications (WISA), 2024.

Domestic Journals & Conferences

1. D. Kang, H. Kim, and S. Hong, "Epoch-based re-keying for full disk encryption to avoid cumulative side-channel leakage," in *Proceedings of the 2026 Summer Conference of the Korea Institute of Information Security & Cryptology (KIISC)*, (in Korean), 2026.
2. D. Kang, I. Lee, H. Kim, and S. Hong, "Side-channel attack on tweakable block cipher SCARF for cache address randomization," in *Proceedings of the 2025 Winter Conference of the Korea Institute of Information Security & Cryptology (KIISC)*, (in Korean), 2025.
3. D. Kang, H. Shin, and S. Hong, "Full-round linear attacks for block cipher DNA-PRESENT," in *Proceedings of the 2025 Summer Conference of the Korea Institute of Information Security & Cryptology (KIISC)*, (in Korean), 2025.

*Earlier publications may appear under the name Dong-U Kang or Dong-u Kang.

Skills

Languages Korean as native language, Strong reading, writing and speaking competencies for English, Japanese.

Coding Python, C, Chipwhisperer, LaTeX, …

Misc. Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

2026 **Special Award**, Korean Cryptographic Contest, Cryptographic Forum, Korea Institute of Information Security & Cryptology (KIISC).

2025 **Encouragement Award**, Cryptographic Analysis Contest, Korea Institute of Information Security & Cryptology (KIISC).

Other Adventures

2026–present **Participant**, Cybersecurity Global Education Programme@Kellogg College, Oxford University.

2024 **Technical Intern**, LIG NEX1.

2023–2024 **Club Leader**, Hoyoung, monochrome photography club@Korea University.

2021–2022 **Staff Photographer, Photo Editor**, Korea University Weekly@Korea University.

References

Available on Request